

Block-Periodic Transverse Modes Obstruct Volume-Uniform Critical Coercivity in a Flat Lattice Gauge Block Form

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Abstract

Let the fine periodic lattice have side LN' and let Q_L be the L^{-d} -normalized block average of length- L line integrals. In four dimensions, the rescaling $Q_L \mapsto LQ_L$ exactly removes the elementary constant-field scaling obstruction to a volume-uniform Poincaré estimate. We prove that it does not remove the full-space obstruction. For every positive coarse side N' , every even block side $L = 2M$, every pair of distinct directions $i \neq j$, and every internal vector w , we construct a block-periodic transverse one-cochain $A_{i,j,w}$. It satisfies

$$Q_L A_{i,j,w} = 0, \quad \text{div } A_{i,j,w} = 0,$$

and its flat Hodge energy and norm are exactly

$$\langle A_{i,j,w}, K_0 A_{i,j,w} \rangle = 8N'(LN')^{d-1} \|w\|^2, \quad \|A_{i,j,w}\|^2 = (LN')^d \|w\|^2.$$

Thus the Rayleigh quotient of $K_0 + (LQ_L)^*(LQ_L)$ on this family is $8/L$, independently of N' . For $N_c \geq 2$, the current constant-before-volume critical coercivity gate is therefore false on every positive coarse torus. More generally, no scalar rescaling $s_L Q_L$ can control this family, because it lies in the exact kernel of Q_L .

The construction, energy count, and quantified no-go theorem are formalized in Lean 4 against pinned Mathlib. A focused six-declaration axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`; there are no project axioms or `sorry`. The result does not disprove repaired sector or quotient estimates and does not address an interacting Hessian, an infinite-volume limit, a continuum limit, or a Yang–Mills mass gap.

1 Introduction

Finite-range coercive operators admit Combes–Thomas decay estimates whose rates are controlled by the lower spectral edge [3]. In a block-spin treatment of lattice gauge fields, a natural candidate precision is a flat Hodge operator plus a positive block penalty. At a fixed volume this is a standard finite-dimensional mechanism. The difficult question is whether its coercivity constant survives uniformly as the block side tends to infinity.

The block map considered here is the linearized line-integral average used in lattice gauge renormalization [1, 2]. With its unscaled normalization, constant one-cochains force a growing Poincaré constant in dimension four. Multiplying the map by the critical factor L repairs that specific scaling: on direction-wise constant fields, LQ_L is isometric at the level relevant to the four-dimensional norm count. This makes the following proposal mathematically natural:

$$\|A\|^2 \leq C_P \left(\langle A, K_0 A \rangle + \|LQ_L A\|^2 \right), \quad C_P \text{ independent of } L. \quad (1)$$

The purpose of this paper is to settle that proposal on its stated full domain. The answer is negative for a structural reason. The block map has transverse low-energy modes in its exact kernel. Repeating a mean-zero square profile within every coarse block makes the cancellation local to each block, so the obstruction persists for every positive coarse side N' , rather than only for the one-site coarse torus. Its energy is carried by two interfaces per block, while its norm is carried by the entire volume; the resulting quotient is exactly $8/L$.

Three features distinguish the result from a heuristic low-momentum argument.

- (i) The witness is an explicit periodic cochain on the exact finite lattice consumed by the block map.
- (ii) Both kernel identities and the Hodge energy are exact, including all finite-volume factors.
- (iii) The constant is quantified before the volume, and the contradiction is formalized at that quantifier order.

The theorem closes the present full-space flat route. It does not close the physically different alternatives of passing to a justified sector or quotient, or adding a positive interacting term that lifts the transverse family. That distinction is essential: a no-go theorem is useful precisely when it identifies which input must change.

2 Finite periodic lattice and the critical block form

2.1 One-cochains and the flat Hodge operator

Fix integers $d \geq 2$, $L \geq 1$, $N' \geq 1$, and $N_c \geq 1$. The fine site set is

$$\Lambda_{L,N'} = (\mathbb{Z}/(LN')\mathbb{Z})^d,$$

and a positive bond is a pair (x, μ) with $x \in \Lambda_{L,N'}$ and $\mu \in \{0, \dots, d-1\}$. The internal coordinate space is

$$E_{N_c} = \mathbb{R}^{N_c^{d-1}},$$

the coordinate model used by the formalization for $\mathfrak{su}(N_c)$. A physical one-cochain is a function $A(x, \mu) \in E_{N_c}$ with the finite ℓ^2 norm

$$\|A\|^2 = \sum_{x \in \Lambda_{L,N'}} \sum_{\mu=0}^{d-1} \|A(x, \mu)\|^2.$$

At the trivial gauge background, let D_1 be the discrete covariant exterior derivative on ordered plaquettes and let δ be the gauge constraint (discrete divergence). The flat Hodge operator K_0 is normalized so that

$$\langle A, K_0 A \rangle = \|D_1 A\|^2 + \|\delta A\|^2. \tag{2}$$

Only this quadratic-form identity is needed below. The formal development constructs D_1 , δ , and K_0 as continuous linear maps on the finite Hilbert spaces, rather than postulating (2).

2.2 The line-integral block map

Write B_y for the fine block associated with a coarse site $y \in (\mathbb{Z}/N'\mathbb{Z})^d$. It contains the sites

$$x = Ly + r, \quad r \in \{0, \dots, L-1\}^d.$$

For a coarse positive bond (y, μ) , the linearized block map is

$$(Q_L A)(y, \mu) = L^{-d} \sum_{x \in B_y} \sum_{k=0}^{L-1} A(x + ke_\mu, \mu). \quad (3)$$

Thus Q_L averages, over the source block, the length- L straight line integral in the coarse bond direction. The critical four-dimensional rescaling is

$$\tilde{Q}_L = LQ_L.$$

For a scalar sequence s_L , set more generally

$$H_{L,s} = K_0 + (s_L Q_L)^* (s_L Q_L). \quad (4)$$

Definition 2.1 (critical full-space gate). In $d = 4$, for fixed N' and N_c , the critical gate asserts that there is a number $C_P > 0$, chosen independently of L , such that

$$\|A\|^2 \leq C_P \left(\langle A, K_0 A \rangle + \|LQ_L A\|^2 \right) \quad (5)$$

for every positive block side L and every fine one-cochain A .

The quantifier order is part of the claim. At any fixed L the problem is finite-dimensional, whereas Definition 2.1 asks one constant to work before all volumes are chosen.

3 A block-periodic transverse sector

Let $L = 2M$ with $M \geq 1$. Define the square sign on the within-block cycle by

$$\sigma_M(r) = \begin{cases} +1, & 0 \leq r < M, \\ -1, & M \leq r < 2M, \end{cases} \quad r \in \mathbb{Z}/(2M)\mathbb{Z}, \quad (6)$$

and extend it to the full fine cycle by block periodicity:

$$\sigma_{M,N'}(m) = \sigma_M(m \bmod 2M), \quad m \in \mathbb{Z}/(2MN')\mathbb{Z}.$$

The two elementary identities are

$$\sigma_{M,N'}(m)^2 = 1, \quad \sum_{r=0}^{2M-1} \sigma_M(r) = 0. \quad (7)$$

Choose distinct directions $i \neq j$ and $w \in E_{N_c}$. Define the one-cochain

$$A_{i,j,w}(x, \mu) = \begin{cases} \sigma_{M,N'}(x_j)w, & \mu = i, \\ 0, & \mu \neq i. \end{cases} \quad (8)$$

The field points in direction i and varies only in the transverse direction j . For a fixed phase of the square wave, the collection

$$\mathcal{T}_{M,N'} = \text{span}\{A_{i,j,w} : i \neq j, w \in E_{N_c}\}$$

is the explicit obstructive family used in this paper. We do not claim that $\mathcal{T}_{M,N'}$ is the entire kernel or the entire low-energy spectral subspace.

Lemma 3.1 (exact norm). *For every $i \neq j$ and $w \in E_{N_c}$,*

$$\|A_{i,j,w}\|^2 = (LN')^d \|w\|^2. \quad (9)$$

Proof. Only bonds in direction i contribute. There is one such bond at each of the $(LN')^d$ fine sites, and every coefficient has squared magnitude one by (7). $\square$

Lemma 3.2 (blockwise kernel identity). *For every positive coarse side N' ,*

$$Q_L A_{i,j,w} = 0. \quad (10)$$

Consequently $(s_L Q_L) A_{i,j,w} = 0$ for every scalar s_L .

Proof. If the coarse bond direction is not i , every summand in (3) vanishes. In direction i , shifting along the line does not change the transverse coordinate x_j . The line sum therefore contributes a factor L , but the sum over the j -offset inside the source block contains

$$\sum_{r_j=0}^{L-1} \sigma_M(r_j) = 0.$$

This cancellation occurs separately in every source block. It is therefore independent of the number N' of blocks along each direction. $\square$

Lemma 3.3 (transverse divergence). *The witness is divergence-free at the trivial background:*

$$\delta A_{i,j,w} = 0. \quad (11)$$

Proof. Only the i component can enter the divergence. That component is unchanged by a forward or backward shift in direction i , because it depends only on x_j and $i \neq j$. $\square$

The preceding two lemmas locate the obstruction: it lies simultaneously in the block-map kernel and the flat gauge-constraint kernel. Its remaining energy is purely curvature energy.

4 Exact interface count and Rayleigh quotient

Let S be the successor map on the full fine cycle of length $LN' = 2MN'$. There are two sign changes in each block, including the boundary between the end of one block and the start of the next. At a sign change the difference has magnitude 2 and squared magnitude 4. Hence

$$\sum_{m \in \mathbb{Z}/(LN')\mathbb{Z}} (\sigma_{M,N'}(m) - \sigma_{M,N'}(Sm))^2 = 8N'. \quad (12)$$

Lemma 4.1 (finite-volume interface count). *For any direction j ,*

$$\begin{aligned} \sum_{x \in \Lambda_{L,N'}} (\sigma_{M,N'}(x_j) - \sigma_{M,N'}((x + e_j)_j))^2 \\ = 8N'(LN')^{d-1}. \end{aligned} \quad (13)$$

Proof. Split the finite box into its j coordinate and the remaining $d - 1$ coordinates. Equation (12) gives the first factor; the remaining coordinates give $(LN')^{d-1}$ identical copies. $\square$

Proposition 4.2 (exact flat Hodge energy). *For $i \neq j$,*

$$\langle A_{i,j,w}, K_0 A_{i,j,w} \rangle = 8N'(LN')^{d-1} \|w\|^2. \quad (14)$$

Proof. By Lemma 3.3, the divergence term in (2) vanishes. In the plaquette curl, every ordered direction pair except the pair containing i and j cancels: either the field component is zero, or the transverse profile is unchanged by the shift. On the (i, j) plaquettes, the squared norm is exactly the scalar square difference in Lemma 4.1 times $\|w\|^2$. The two possible orderings of i and j give the same value under the convention that each geometric plaquette is counted once. This proves (14). $\square$

Theorem 4.3 (exact Rayleigh quotient). *If $w \neq 0$, then for every scalar s_L ,*

$$\frac{\langle A_{i,j,w}, H_{L,s} A_{i,j,w} \rangle}{\|A_{i,j,w}\|^2} = \frac{8}{L}. \quad (15)$$

In particular, if c_L denotes the best coercivity constant of $H_{L,s}$ on the full one-cochain space, then

$$c_L \leq \frac{8}{L}. \quad (16)$$

Proof. Lemma 3.2 removes the block term in (4). Divide Proposition 4.2 by Lemma 3.1 and cancel the positive factors N' , $(LN')^{d-1}$, and $\|w\|^2$. $\square$

The cancellation of N' is the central finite-volume point. The obstruction is neither diluted nor repaired by enlarging the coarse torus. It is repeated block by block, and the interface and norm counts scale compatibly.

5 No volume-uniform critical coercivity

We now specialize to physical dimension $d = 4$. The internal space is nonzero when $N_c \geq 2$, since $N_c^2 - 1 > 0$.

Theorem 5.1 (all-coarse-volume critical no-go). *Fix $N' \geq 1$, $N_c \geq 2$, and any adjoint-model parameter used to define the flat operators. There is no $C_P > 0$ such that*

$$\|A\|^2 \leq C_P \left(\langle A, K_0 A \rangle + \|LQ_L A\|^2 \right) \quad (17)$$

holds for every block side $L \geq 1$ and every fine physical one-cochain A . Equivalently, the proposition

$$\text{VolumeUniformCriticalRescaledFlatPoincareGate } N' N_c \rho$$

is false for every positive N' .

Proof. Assume that such a C_P exists. Choose an even block side $L = 2M$ and a nonzero internal vector w . Substituting $A_{i,j,w}$ into (17), using Lemma 3.2, and applying (9) and (14) gives

$$(LN')^4 \|w\|^2 \leq C_P 8N' (LN')^3 \|w\|^2.$$

All cancelled quantities are positive, so

$$L \leq 8C_P. \tag{18}$$

The asserted gate supplies the same C_P before all volumes are chosen. Choose M so that $2M > 8C_P$. This contradicts (18). $\square$

Corollary 5.2 (Poincaré constants diverge). *At each even scale, every valid full-space constant satisfies*

$$C_P(L) \geq \frac{L}{8}.$$

This lower bound is independent of N' .

Corollary 5.3 (scalar rescaling cannot repair the gate). *Replacing LQ_L by $s_L Q_L$, for an arbitrary real scalar sequence s_L , does not produce full-space volume-uniform coercivity.*

Proof. The same witness satisfies $(s_L Q_L)A_{i,j,w} = 0$ exactly, so Theorem 4.3 and the contradiction are unchanged. $\square$

5.1 Consequence for localization estimates

Suppose a covariance-root estimate uses a coercivity input c_L and has an amplitude bounded below by the usual spectral scale $2/\sqrt{c_L}$. The witness forces

$$\frac{2}{\sqrt{c_L}} \geq \sqrt{\frac{L}{2}}.$$

Thus no bound built from this full-space coercivity input can have a volume-uniform root amplitude. This statement identifies the failing parameter; it does not invalidate separate finite-range, kernel-amplitude, ball-count, or tilt-budget estimates.

6 What the theorem closes, and what it leaves open

Theorem 5.1 is a domain diagnosis. It rules out exactly the full one-cochain space equipped with the flat Hodge term and a scalar multiple of the present block map. Several logically different continuations remain.

Restricted or quotient sector. One may remove or quotient the block-periodic transverse sector, provided the restriction is physically justified and stable under the operations required by the renormalization step. Merely declaring the witness absent is not enough: the repaired domain must be defined, its invariance proved, and its coercivity established with a constant uniform in L .

Additional positive Hessian. An interacting Wilson Hessian, or another positive term V_L , could lift the family if it satisfies

$$\langle A_{i,j,w}, V_L A_{i,j,w} \rangle \geq c_* \|A_{i,j,w}\|^2$$

with $c_* > 0$ independent of L and N' . Such a statement is outside the flat calculation and is not assumed here.

Modified block operator. A non-scalar modification could in principle see the transverse family. The corollary above shows why changing only the scalar normalization cannot: the problem is the exact kernel, not the size of a nonzero singular value.

No continuum claim. The argument is finite-dimensional and exact. It proves neither existence of an infinite-volume measure nor a continuum limit, and it supplies no positive Yang–Mills mass gap. Its role is to prevent a false uniform input from entering those later arguments.

7 Machine-checked realization

The formalization, carried out in Lean 4 and Mathlib [4, 5], is contained in

`YangMills/RG/PhysicalCriticalRescalingNoGoAllCoarse.lean`.

It imports the previously checked one-coarse-site audit and proves the all-coarse-volume result without adding axioms. Table 1 maps the mathematical chain to selected Lean declarations.

Mathematical content	Lean declaration
Block/offset decomposition	<code>blockOf_eq_image_blockOffsetSite</code>
Mean-zero profile in every block	<code>sum_blockPeriodicSquareSign_block_eq_zero</code>
Exact cochain norm	<code>norm_sq_blockPeriodicSquareModeCochain</code>
Exact block-map kernel	<code>flatBlockConstraintQCLM_blockPeriodicSquareMode_eq_zero</code>
Kernel after critical rescaling	<code>criticalScaledBlockConstraintQCLM_blockPeriodicSquareMode_eq_zero</code>
Zero transverse divergence	<code>gaugeConstraintQCLM_blockPeriodicSquareModeCochain_eq_zero_of_ne</code>
Exact curvature norm	<code>norm_sq_covariantD1CLM_blockPeriodicSquareModeCochain_of_ne</code>
Exact Hodge quadratic form	<code>flatGaugeHodgeK0_inner_blockPeriodicSquareModeCochain_of_ne</code>
All- N' no-go theorem	<code>volumeUniformCriticalRescaledFlatPoincareGate_false</code>

Table 1: Theorem–artifact map.

The formal proof mirrors the analytic proof but makes four finite-indexing steps explicit: the equivalence between block sites and within-block offsets; quotient/remainder coordinates on the full cycle; separation of one box coordinate from the remaining coordinates; and the two orderings of the plaquette direction pair. These are the points at which an informal factor of N' , L , or two is easiest to lose.

The focused oracle is

`papers/poincare-wall/CriticalRescalingNoGoAllCoarseOracle.lean`.

For each of the six headline declarations it reports exactly

`[propext, Classical.choice, Quot.sound]`,

the standard logical dependencies expected from the surrounding finite Hilbert-space development. In particular, the audit contains no `sorryAx` and no project-specific axiom.

Reproduction

The checked workspace uses Lean v4.29.0-rc6 and Mathlib revision 076427204801 (full hash pinned in `lake-manifest.json`). From the repository root:

```
lake build YangMills.RG.PhysicalCriticalRescalingNoGoAllCoarse
lake env lean papers/poincare-wall/\
  CriticalRescalingNoGoAllCoarseOracle.lean
lake build YangMillsCore
```

The verified runs completed with 8178 jobs for the focused module and 8475 jobs for the full core. The source file itself also passes direct Lean elaboration. Warnings emitted by the full core are pre-existing linter warnings and do not affect theorem checking.

8 Conclusion

Critical rescaling repairs the constant-field normalization but cannot repair an exact kernel. The block-periodic transverse family constructed here lies in that kernel on every positive coarse torus, is divergence-free, and has only $O(L^{-1})$ flat Hodge energy per unit norm. Therefore the current full-space critical block/Hodge form has no volume-uniform positive lower bound, for any scalar normalization of the block map.

The conclusion is negative but operational: the localization machinery is not the failing component. The lower spectral edge is. Any successful continuation must alter the physical domain or add an operator that gives uniform energy to the explicitly identified transverse sector.

References

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